

Computer Science & Information Technology

C Programming

Control Flow Statements

DPP: 1

Q1 #include <stdio.h>

int main(void)

{

int i = 2, j = 3, k = 4;

if (i < j ? 1 : 0)

printf("GATE");

else

printf("Wallah2023");

return 0;

}

The output of the program is ____.

(A) GATE

(B) 2023WALLAH

(C) TEWALLAH

(D) EWALLAH

Q2 #include <stdio.h>

void main()

{

int a, b, c, d;

a = 2; b = -1; c = 3; d = -4;

if(a = b - c - d)

printf("%d%d%d", a++, b--, c++);

else

printf("%d%d%d", c--, ++a, ++b);

}

The output is ____.

(A) 1-2 4

(B) 3 1 0

(C) 2 1-3

(D) 3 3 0

Q3 #include <stdio.h>

int main(void)

{

int a = 3 > 2 ? 0 : 0 : 1 : 5;

if(a == a - 1)

printf("GATE 2023");

else

printf("GATE WALLAH");

return 0;

}

The output of the program is ____.

(A) GATE 2023

(B) GATE WALLAH

(C) Compiler error

(D) Garbage value

Q4 #include <stdio.h>

void main()

{

int a;

a = printf("GATE Wallah 2023");

if(a%4 == 0)

a = a + 5;

else

a = a - 5;

printf("%d", a++);

}

The value of a at the end of the program is ____.

Q5 #include <stdio.h>

void main()

{

int i, j, k;

j = 4;

k = 0;

i = j < k ? k : j --;

if (j < i)

j = j + k - 1;

if (j == i)

j = j - i;

else

j = j + -- k;

printf ("%d", j + k - i);

}

The output is ____.

Q6 Consider the following program:



```
#include<stdio.h>
int main()
{
int a=19, b=20;
if(a++<b--) printf("%d",a+++--b);
else printf("%d", ++a+--b);
return 0;
}
```

The output is 38.

Q7 #include<stdio.h>
void main()
{
int a=0;
printf("%d", a);
if(a=2){
printf("Hi");
printf("%d",a);
}else{
printf("Bye");

```
}
printf("%d", a);
}
```

The output string is:

- (A) 0Hi22 (B) 0Hi20
(C) 0Bye0 (D) 0Hi00

Q8 #include<stdio.h>
void main()

```
{
int a=0, b=0;
a=(a=4)||b=1;
if(a&&b) printf("CProgramming");
else printf("PhysicsWalah");
printf("%d",b);
}
```

The output is-

- (A) CProgramming0 (B) CProgramming1
(C) PhysicsWalah0 (D) PhysicsWalah1



Answer Key

Q1 (A)
Q2 (B)
Q3 (B)
Q4 22~22

Q5 -4~-4
Q6 38~38
Q7 (A)
Q8 (C)



Hints & Solutions

Q1 Text Solution:

$i < j ? 1 : 0$

In the above expression i value is less than j value, hence it will return 1.

So, it will print GATE.

Q2 Text Solution:

$a = -1 - 3 + 4$

$a = 0$

$a \& 0$

Assignment operator assigns and returns the value

$b++ + b: -1 \ 0$

$a++ + a: 0 \ 1$

$c-- -- : 3 \ 2$

↓

Post decrement (It will print 3, then update to 2)

Output: 3 1 0.

Q3 Text Solution:

$a = \frac{3 > 2}{\text{True}} ? \boxed{0 \ ? \ 0 : 1} : 5$

$a = 1$

Assignment operator assigns the value and returns it.

if ($a = a -$

Output: GATE WALLAH

Q4 Text Solution:

GATE Wallah 2023

↓

$a = 16$

printf returns the number of characters successfully printed

$16 \% 4 = 0 \rightarrow \text{True}$

↓

$a = a + 5$

$a \ 16 \ 21 \ 22$

Hence the final value of a is 22.

Q5 Text Solution:

$\begin{matrix} i & j & k \\ 4 & 4 & 0 \\ & 3 & 2 & 1 \end{matrix} - 1$

$3 < 4$

$j = 3 + 0 - 1 = 2$

$2 \neq 4$

$j = j - 1$

$= 2 - 1$

$j = 1$

printed value = $j + k - i$

$= 1 - 1 - 4$

$= -4$

Q6 Text Solution:

If ($19 < 20$) $\rightarrow$ Condition is true. After the condition is

evaluated, a is incremented to 20 and b is decremented to 19.

Now, $\text{printf}("%d", a+++--b);$ is evaluated. b is decremented to 18. So, $(20+18)$ i.e. 38 is printed.

After

that, a is incremented to 21.

Hence, output is 38.

Q7 Text Solution:

void main()

{

int $a=0$;

printf("%d", a); // 0 is printed

if ($a=2$) { // Assignment operator assigns and returns the assigned value; So 2 is assigned to a and 2 is returned. Any non-zero value is considered true.

printf("Hi"); // "Hi" is printed

printf("%d", a); // Since a contains 2, 2 is printed.

} else {

printf("Bye");

}



```
printf("%d", a); //Since a contains 2, 2 is  
printed  
}
```

Output: 0Hi22

Q8 Text Solution:

a = 0. b=0;

a=(a=4)||(b=1) //Assignment operator assigns and
returns the assigned value. Here, short-circuiting

will be applied. Since the logical operator is OR, if
the first part is true, second part is not evaluated
at all. Hence, b=0, a=1.

if(a && b)//The condition evaluates to 1 && 0 i.e.
0. Hence, else part is evaluated.

Output: PhysicsWalah0



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